

TECHNICAL REPORT TR-2026-03

Silent-Failure Study · Week 1

The Drake Engine Adapter

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25 August 2026

Abstract

This report documents the third and final engine column: Drake 1.56.0, driven through `MultibodyPlant`. It records the installation, the modelling of the N -link chain as a rigid-body mechanism, and validation against the library-independent reference, which Drake matches to 7.2×10^{-14} across $N = 1 \dots 5$. All three engines now agree. Two findings are recorded: a coordinate-convention error of the same class as the one in TR-2026-01, caught by the same diagnostic; and the conclusion that only one of the three nominally portable axes transcribes directly into a rigid-body engine, which is a scope question the study must settle before Week 2.

1 Installation

Drake was installed into a virtual environment under WSL Ubuntu 24.04.3 LTS (Python 3.12.3, x86_64):

```
python3 -m venv ~/drake-venv
~/drake-venv/bin/pip install drake numpy scipy sympy
```

Result: `drake 1.56.0`. No `sudo` was required, since a virtual environment sidesteps the PEP 668 restriction that Ubuntu 24.04 places on system-wide `pip` installation.

1.1 The execution boundary

Drake runs on Ubuntu; the rest of the harness runs on native Windows. The adapter is therefore written to stand alone — it imports only `numpy`, `pydrake`, and `reference`, and emits JSON for the Windows side to merge. This works because `reference.py` needs nothing but `numpy` in its core path, so the same referee adjudicates Drake as adjudicates the other two engines, on both sides of the boundary.

2 Modelling

The suite's chain is N point masses on massless rods. As a Drake mechanism:

- Body i 's frame origin sits at joint i ; its point mass sits at $(0, 0, -l)$ in that frame, at the far end of link i .
- Joint $i + 1$ is placed at $(0, 0, -l)$ in body i 's frame — the position of mass i — so each link pivots about the previous mass.
- The revolute axis is $+y$ and gravity is $-z$, so a link hangs along $-z$ at zero angle, matching the suite's convention of measuring from the downward vertical.

Two API details are worth recording, as both cost time. Drake 1.56 has removed `SpatialInertia.PointMass`; the current equivalent is `MakeFromCentralInertia` with a zero `RotationalInertia`. And `RevoluteJoint` requires its two frames to be coincident, so the pivot offset cannot be passed to the constructor — each joint past the first needs an explicit `FixedOffsetFrame` on the parent body.

Drake exposes the mass matrix, the bias term, and generalised gravity separately, so the adapter solves

$$M(q) a = \tau_g(q) - C(q, v) v, \quad (1)$$

which carries the same content as the reference's $Ma = -(C + G)$.

3 A second convention trap

The first validation showed Drake disagreeing with the reference by 2.5 to 6.4 relative, **with the error growing in N** , while passing exactly at $N = 1$. This is the second occurrence of the same failure shape recorded in TR-2026-01 §5, and it was again a defect in the comparison rather than in the engine.

The suite’s Lagrangian uses **absolute** angles measured from the downward vertical — its kinetic coupling $\cos(\theta_j - \theta_k)$ is only meaningful for absolute angles. A chain of Drake revolute joints reports **relative** angles: joint i rotates its body with respect to its *parent*, not with respect to the world. The two coincide for the first link and diverge increasingly thereafter.

The map is linear, constant, and lower triangular,

$$q_0 = \theta_0, \quad q_i = \theta_i - \theta_{i-1} \quad \Longleftrightarrow \quad \theta = \text{cumsum}(q), \quad (2)$$

and because it is constant the same relation carries the velocities and accelerations. Applying it on the way in and out reduced the worst mismatch from 6.4 to 7.2×10^{-14} .

The diagnostic generalises. Two of the three adapters harboured a convention mismatch invisible at the smallest system size — the canonical (q, p) state in one case, relative joint angles in the other. In both, the tell was identical: exact agreement at $N = 1$, growing disagreement beyond. A genuine derivation error rarely switches on at a particular system size; a convention mismatch characteristically does, because degenerate cases make differing conventions coincide. *Any* apparent cross-engine disagreement should be tested against this pattern before it is reported as a finding.

4 Validation

Sixteen pseudorandom states per case, seed 20260822, tolerance 10^{-8} relative.

N	Status	vs. reference
1	pass	0.00
2	pass	2.05×10^{-15}
3	pass	1.07×10^{-14}
4	pass	3.83×10^{-14}
5	pass	7.21×10^{-14}

Drake’s residual grows gently with N — roughly linearly — consistent with accumulation in a recursive rigid-body algorithm, against the reference’s direct closed form. At 7×10^{-14} it is six orders of magnitude inside tolerance and carries no consequence for adjudication.

5 Portability: only one axis transcribes

Three axes were classified as portable in TR-2026-01. Building the Drake column shows that classification was too generous.

Axis	Status in a rigid-body engine
dof	Directly portable. An N -link chain is a mechanism, and all three engines express it natively.
mass_ratio	Expressible as prismatic joints with force elements, but it is a spring network rather than a mechanism. The translation is a modelling choice rather than a transcription.
near_singular	Requires the mass matrix $\begin{bmatrix} m & cm \\ cm & m \end{bmatrix}$. A rigid-body engine yields a diagonal mass matrix for two independent prismatic degrees of freedom; the off-diagonal term can only be induced geometrically, via two prismatic axes at angle $\arccos c$ separated by a <i>massless</i> intermediate body. That is expressible but contrived — and a massless body is itself the class of degeneracy under study, so the construction would change what the axis tests.

The consequence is that dof may be the study’s only genuine three-way axis — ten of the fifty-five cases. This must be settled before Week 2, because it determines the width of the cross-engine claim.

Three responses are available, and the choice is the study’s to make. Restrict the cross-engine claim to dof and report the remaining axes as single-engine characterisation. Or re-express the two spring systems on the pendulum chain, which is what SCOPE.md describes in the first place (TR-2026-01 §7.1) and would make them mechanisms rather than abstract Lagrangians. Or admit the contrived Drake constructions and disclose that the translation is not faithful. The second is the most defensible and the most work.

6 Status

	MechanicsDSL	SymPy	Drake
Adapter	native	✓	✓
Agrees with reference	yes	yes	yes
Worst residual, dof	3.6×10^{-15}	6.1×10^{-15}	7.2×10^{-14}
Known scaling wall	Hamiltonian $N=3$	none at $N \leq 5$	none at $N \leq 5$

All three columns exist, and all three agree. The Week 1 objective — the same system expressed three ways, agreeing on a case checkable by hand — is met.

What has not been tested. Every result so far compares *derivations*: each engine is asked for its equations of motion and checked against the reference. None of it integrates anything. The ten timeouts in the frozen baseline all occur during simulation, so the half of the pipeline where the known failures live remains unexercised across engines. That is the next measurement, and the likeliest place for a genuine disagreement to appear.

Artefacts. stress_suite/adaptor_drake.py, stress_suite/out/drake.json. Committed at 47b29bb.

Disclosure. The author maintains one of the engines measured. AI assistance was used in this work. All figures were produced by executing the committed code.